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Machine Learning Approach to Forecast Work Zone Mobility using Probe Vehicle Data

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Abstract

The aim of deploying intelligent transportation systems (ITS) is often to help engineers and operators identify traffic congestion. The future of ITS-based traffic management is the prediction of traffic conditions using ubiquitous data sources. There are currently well-developed prediction models for recurrent traffic congestion such as during peak hour. However, there is a need to predict traffic congestion resulting from non-recurring events such as highway lane closures. As agencies begin to understand the value of collecting work zone data, rich data sets will emerge consisting of historical work zone information. In the era of big data, rich mobility data sources are becoming available that enable the application of machine learning to predict mobility for work zones. The purpose of this study is to utilize historical lane closure information with supervised machine learning algorithms to forecast spatio-temporal mobility for future lane closures. Various traffic data sources were collected from 1,160 work zones on Michigan interstates between 2014 and 2017. This study uses probe vehicle data to retrieve a mobility profile for these historical observations, and uses these profiles to apply random forest, XGBoost, and artificial neural network (ANN) classification algorithms. The mobility prediction results showed that the ANN model outperformed the other models by reaching up to 85% accuracy. The objective of this research was to show that machine learning algorithms can be used to capture patterns for non-recurrent traffic congestion even when hourly traffic volume is not available.

Recurrent traffic congestion is experienced daily by highway users worldwide as a result of over-capacity traffic conditions during peak hours. Non-recurrent traffic congestion occurs irregularly as a result of road construction projects, traffic incidents, and inclement weather. Highway construction projects often require lane or shoulder closures to provide an area for construction. The reduction in the number of available lanes reduces highway capacity which leads to congestion when the roadway supply is outweighed by the traffic demand. These traffic jams propagate into upstream segments which increases the chance of rear-end traffic incidents. In the U.S.A., these lane closures constituted almost one quarter of the total non-recurrent traffic congestion and caused more than 96,000 crashes in 2014 (*1*). In addition to these mobility and safety concerns, negative environmental impacts, waste of fuel, emissions, and user dissatisfaction are also caused by the presence of work zones.

Transportation agencies have been using various modeling approaches to predict the impact that these lane closures have on traffic mobility to prepare appropriate traffic management plans to minimize negative impacts. Among these approaches, deterministic queuing theories (*2–4*) and traffic simulation software have been

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widely adopted to determine traffic delays and queue lengths (*5*, *6*). The deterministic queuing models consider a relationship between available highway capacity and approaching traffic volume to forecast the mobility impact. Traffic simulation software has also been used to create and evaluate a synthetic model of the real-world lane closure scenarios (*7*). Although these approaches have sufficed in the past, they require accurate hourly traffic volume counts as a critical input, which are often unavailable, unreliable, or expensive.

Traffic congestion prediction is a vital part of intelligent transportation systems (ITS) which provide traffic mobility information for both road users and transportation agencies. ITS help traffic operation centers (TOCs) and state departments of transportation (DOTs) to design their traffic management plans proactively to minimize the mobility and safety concerns related to lane closures. In the era of big data, machine learning algorithms are capable of learning dynamic patterns from previous real-world examples and predicting future scenarios. These algorithms explore historical observations to learn underlying congestion patterns and predict future scenarios. Although these algorithms have been widely used to predict mobility during recurring traffic congestion experienced in peak hours, there have been limited research efforts to examine their applicability to forecast mobility during non-recurrent traffic congestion resulted from work zone presence. Currently, traffic mobility data from these historical work zones is available which provides this opportunity to train machine learning models and predict mobility for future work zones. This study seeks to train supervised learning algorithms using historical work zone mobility profiles and forecast mobility, or the speed for each segment over time, for future lane closures. Despite previous studies that included hourly traffic volume as a critical input, this study only utilizes the distribution of historical traffic speed to provide a mobility baseline for each segment.

Background

Significant past efforts have been made to characterize the impacts of lane closures on traffic mobility using parametric and simulation approaches. Only recently have there been limited research efforts to forecast mobility for these lane closures using machine learning techniques. It is important to understand these efforts and utilize available data sources to improve current practices.

Prediction of Work Zone Traffic Mobility

Traditionally, traffic engineers use parametric approaches such as deterministic queuing theory (*2*, *4*, *8*) and shock-wave theory (*9*, *10*) to estimate traffic mobility in work zones. For decades, these approaches have been widely used to assist state DOTs and TOCs to understand the impact that lane closures have on measures of traffic mobility such as travel time and queue length. Both of these methods use hourly traffic volume and predicted capacity as critical inputs to forecast traffic delay when a lane closure reduces the available capacity. Although deterministic queuing theory provides a decent delay estimation, it does not suit situations when more accurate spatio-temporal traffic behavior prediction is required (*11*). Another fundamental and commonly used method is based on shock-wave theory, which uses fluid flow behavior to predict traffic flow in a spatio-temporal manner (*9*, *10*). In this method, a flow-speed-density relationship is used to determine traffic flow transitions based on the traffic volumes and available capacity. Traffic simulation software such as VISSIM (*12*) and Quickzone (*13*) also attempt to simulate this capacity reduction scenario to determine traffic flow behavior. The Highway Capacity Manual (HCM) (*14*) also provides a formula to calculate highway capacity and assess traffic flow. In addition to parametric and simulation approaches, machine learning algorithms have been applied to address limitations associated with the previously mentioned approaches. For instance, artificial neural network (ANN) has been used to predict queue length and delay measures (*15–17*), as well as capacity reduction (*18*) resulting from lane closures.

Probe Vehicle Data

Probe vehicle data is collected from GPS devices in vehicles traveling on highways. This technology aggregates speed data over specific highway segments termed traffic message channels to provide a representative speed for segments of road over time. Third-party vendors such as INRIX (*19*) and HERE (*20*) provide these data for DOTs, TOCs, and navigation applications which provide real-time traffic conditions and travel time information for commuters. These data were verified by numerous sources including the I-95 Corridor coalition (*21*). Since then, these data have been used in different areas of work zone traffic mobility including impact measurement (*22–24*), congestion and bottle neck evaluation (*25–27*), and traffic mobility prediction (*11*, *24*, *28*).

Although these data have been used extensively in recurrent traffic congestion prediction, they have not been used widely to forecast non-recurrent traffic congestion resulting from highway lane closures. Du et al. used probe vehicle data along with an ANN model for estimating spatio-temporal freeway work zone delay (*11*). Du et al. also forecasted work zone delay and

cost using a hybrid machine learning model consisting of an ANN model with one hidden layer coupled with a support vector machine (SVM) model. The SVM was initially used to predict and feed capacity to the ANN model for traffic speed prediction (*28*). In both studies, hourly traffic volume was used as the critical input to train the predictive models. In the absence of hourly traffic volume, a classification modeling approach was introduced, in an earlier attempt, to predict speed ranges for each highway segment during lane closure events (*24*). Distributions of historical traffic speeds were used to provide a mobility baseline for random forest and XGBoost classification algorithms (*24*). These models showed a decent performance, however, there are various approaches that can potentially improve this prediction methodology. Utilizing historical speed distribution to define the mobility baseline showed promising results, though this approach is still in its nascent stage as there are numerous ways to query and quantify historical speed distribution. In addition, highway mobility data sets suffer from the imbalanced data issue which is inevitable in highway congestion prediction practices. A common approach to address this issue is to use resampling algorithms to create balanced data sets for model training. Also, including additional parameters along with more advanced predictive models could potentially improve prediction accuracy and model generalization. Therefore, this study attempts to extend previous studies by:

- introducing a new approach to define and quantify mobility baseline for each segment over time;
- examining the applicability of different resampling approaches to overcome the imbalanced data issue;
- examining the generalization of this approach in a statewide scale;
- designing an ANN architecture to improve prediction accuracy.

Methodology

This study adopted a supervised machine learning approach to estimate traffic speeds for highway segments when lane closures occur. The severity of the traffic interruptions depends on the number of available lanes and the approaching traffic volume. When there is not enough capacity for traffic, vehicular speeds decrease and congestion propagates to upstream segments. As this congestion propagation continues, the upstream traffic segments experience lower traffic speeds. Using probe vehicle data, these speeds were available to investigate mobility during historical lane closure projects. These traffic speeds ranged from zero miles per hour (mph), when vehicles were stopped in a queue, to greater than 70 mph when traffic was operating in free-flow condition.

A comprehensive way to visualize traffic mobility is to create a matrix of traffic speeds over time and space. Figure 1 shows a traffic speed heat map in which the x-axis shows time and the y-axis represents highway segments. This figure illustrates traffic speeds for each segment over time for a single lane closure on northbound I-75 in Oakland County. This lane closure was in place on a Tuesday in September 2014. Traffic speeds are shown using a spectrum of colors from red to green to represent low speed and high speed records, respectively. The yellow and red areas on the figure show traffic congestion while green areas show that traffic was operating at higher speeds. During morning and off-peak hours, highway segments were serving traffic with high speeds since there was sufficient capacity for approaching traffic volumes. However, traffic congestion happened during evening peak hour because of overwhelming traffic volumes for the remaining two lanes. As highway segments reached their capacity, traffic congestion propagated in the upstream segments and traffic jam stretched up to about 9 mi. From the temporal perspective, the traffic congestion and severe slow down was experienced for 5 h from 3:00 to 8:00 p.m.

Typical Traffic Mobility Baseline

One of the objectives of this study was to use the historical speed distribution as an input to train the machine learning models and predict potential future congestion. Data from probe vehicles provide an opportunity to observe historical trends in traffic and to analyze those trends to understand future congestion risk on a segment of roadway (*29*). Historical speed data from the same day of week, hour of day, and 15-min bin for each segment was queried from the year before the lane closures. Several percentile values were used to represent speed distributions (traffic behavior) for each segment. In statistics, a percentile measure is used to identify a value in a group of observations below which a given percentage of

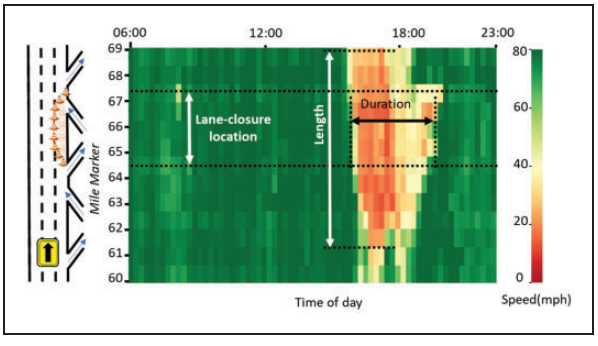


Figure 1. Speed heat map for a single lane closure.

observations can be found. For instance, if the 85th percentile of historical speeds for a highway segment is 65 mph, it means that 85% of traffic speeds observed for this segment are below 65 mph. Conversely, this also means that speeds were above 65 mph for only 15% of cases. The 5th, 15th, 25th, 50th, 75th, 85th, and 95th percentiles were calculated to quantify historical speed distribution.

Figure 2 serves as an illustration that historical speed data can be used to identify where potential issues may occur. Figure 2 shows historical speed profiles at various percentile values for the same work zone introduced earlier. Figure 2*a* shows the 85th percentile speeds for each segment during a typical Tuesday. This corridor efficiently services traffic at the 85th percentile level, except for a very small period of low speeds in the evening peak hour (callout i). Figure 2*b* shows that in the 50th percentile level, the congestion in the evening period is more apparent (callout ii). Figure 2*c* shows the 25th percentile speeds in the evening period beginning to drop below 20 mph (callout iii). Figure 2*d* shows the 5th percentile speeds and congestion become apparent in the evening peak hour (callout iv). These four figures begin to illustrate that there are potential mobility issues in the evening period. Figure 2*e* shows mobility for this corridor when a single lane closure work zone was in place. It is clear that the locations that experienced low speeds historically were affected by the lane closure (callout v).

Figure 2 illustrates the applicability of Greenshield's theory (*30*) in which traffic speed and traffic volume are related. This was, indeed, the motivation behind this study to examine the applicability of using the distribution of historical traffic speeds, when hourly traffic volumes are not available, to train machine learning models. Modern technology provides the opportunity to gather historical mobility data for these work zones and apply machine learning algorithms to learn from these historical observations and predict mobility for future scenarios. This study used a supervised learning approach to predict speed ranges for each highway segment over time. Classification algorithms used in this study were random forest, XGBoost, and ANN. The following section provides a brief explanation of these algorithms and their applicability for classification tasks.

Model Selection

The algorithms chosen for this study were based on their popularity and performance compared with other machine learning algorithms. Decision tree models were chosen because of their capability to visualize their decision making processes using trees. A neural network model was chosen because of its capability to capture non-linear and hidden patterns in a data set. These models are briefly introduced and discussed in the following sections.

Artificial Neural Network. An artificial neural network (ANN) is a computational model based on the structure and functions of biological neural networks (*31*). The first computational model for ANN was created by McCulloch and Pitts (*32*) based on mathematics and algorithms. Neural network models contain at least one hidden layer between input (training data) and output layer (predictions) to transform the inputs into something that the output layer can use. In hidden layers, neurons take in a set of weighted inputs and produce an output through an activation function to minimize the prediction error. ANN models use large number of neurons to identify hidden patterns in previous occurrences. These models are capable of predicting complex scenarios (*31*) even though they are computationally expensive. This study used Scikit-learn library in Python 3.6 to import and apply a multilayer perceptron

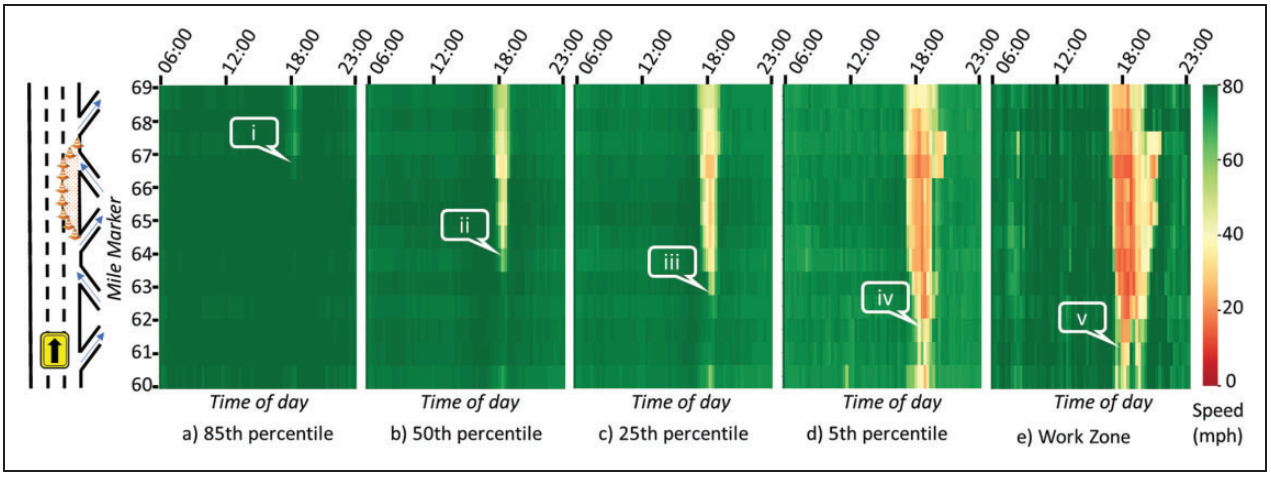


Figure 2. Speed heat maps for historical and lane closure scenarios.

(MLP) classifier. The MLP is a class of feedforward ANN that optimizes the log-loss function using a stochastic gradient descent method (*33*).

Random Forest. Random forest or random decision forest is an ensemble learning method designed to construct multiple decision trees from previous observations and to predict future scenarios (*34*, *35*). The first algorithm for random decision forests was created by Ho (*34*). Single decision trees are prone to over-fitting while random forest models use votes from several decision trees to reduce variance in predictions (*36*). This comes at the expense of a small increase in bias and some loss of interpretability, but generally boosts the performance in the final model. In almost all cases, random forests are more accurate than decision trees but are more computationally expensive (*34*). The random forest algorithm used in this study was imported from the Scikit-learn library in Python 3.6 (*33*).

XGBoost. Extreme gradient boosting (XGBoost) is a decision-tree-based ensemble algorithm that uses a gradient boosting framework technique for regression and classification problems. This algorithm was proposed by Chen (*37*) in 2016, and since then has been used by data scientists in various competitions because of its fast running speed and high performance. The most important factor behind the success of XGBoost is its scalability in all scenarios which is because of several important systems and algorithmic optimizations. These innovations include: a novel tree learning algorithm for handling sparse data; a theoretically justified weighted quantile sketch procedure to handle instance weights in approximate tree learning; and a parallel and distributed computing feature that makes the learning process faster. The XGBoost algorithm used in this study was imported from the XGBoost package in Python 3.6 (*37*).

Data Collection

Data was collected from 1,160 highway lane closure projects on Michigan interstates from 2014 to 2017. These work zones contained both single lane and double lane closures with durations from one to 15 days. Lane closures of less than one day were not considered because there was not enough information to verify the exact times lanes were opened or closed. The physical lengths of the work zones ranged from 100-foot bridge repairs to 7-mi pavement reconstruction. The information provided by the Michigan Lane Closure and Restrictions (LCAR) database did not include work zone configurations, such as barrel placement, taper lengths, and signage. This information would clearly be valuable for future model development.

The probe vehicle data for each work zone was collected between 5 mi upstream of the starting point of the work zone and 3 mi downstream of the ending point of the work zone. Average annual daily traffic (AADT) values were collected for each work zone from the Highway Performance Monitoring System (HPMS) database (*38*). These AADT values were spatially joined with the highway segments to approximate approaching traffic volumes. The Michigan Traffic Crash Facts (MTCF) (*39*) website was also used to query all traffic crashes that occurred during the lane closure periods. The MDOT lane mile inventory was used to collect geometric features of the roadway including number of lanes and functional road class. Historical traffic speeds were obtained using probe vehicle data, which was provided by a third-party vendor. These historical speeds were aggregated into 15-min periods, for each unique day of week and hour of day. For instance, if the lane closure was taking place on a Monday from 1:15 to 1:30 p.m., traffic speeds were queried for all Mondays in the previous year during the same time period (1:15 to 1:30 p.m.). Using this approach, traffic speed distributions could be created for each segment. This aggregation strategy also removed some of the noise in the speed data set to provide more stable historical traffic speeds. A panel data set was constructed for each of the lane closures occurring on different interstate highways in the State of Michigan. Panel data sets are defined as multidimensional data with a time component (*40*). The final data assembled has numerous variables that differ over time, including traffic characteristics such as AADT, geometry, and traffic incidents, as well as lane closure information. The final data set is comprised of over one million records.

Data Preprocessing

The mobility data, aggregated into 15-min bins, were paired with geometric information, temporal features, AADT, and spatial features. The speed records, which were originally continuous speeds between 0 and 80 mph, were converted into four categories: Class 1 (0–20 mph), Class 2 (20–40 mph), Class 3 (40–60 mph), and Class 4 (60–80 mph).Past literature has defined congestion in speed ranges varying from 5 to 25 mph (*41*, *42*). This study used 20 mph speed ranges for the primary goal of identifying high level changes in roadway traffic conditions as a result of work zones. Also, categorical features were converted to binary features using one-hot encoding techniques. The final data set consisted of 83 features for each 15-min speed. Traffic data are typically considered noisy. Sources that contribute to this noise are traffic incidents, inclement weather, lane closures, and unexpected driving behaviors. It was beneficial to clean

problematic data records that could potentially confuse our algorithms. Traffic incidents were cross-checked for each case study. Case studies that experienced a traffic incident were removed from the data set.

Resampling Techniques

After constructing the final data set for a multi-class classification task, the number of records for each class label showed that the data set was highly imbalanced. There were far more high speed records than low speed records since interstate highways typically operate at higher speeds. Figure 3 illustrates the frequency of speed classes for both single lane and double lane case studies.

To address this imbalanced data issue, several resampling techniques were applied using the imbalanced-learn (*43*) library in Python. The techniques used were random under-sampling (*44*), over-sampling (*45*), and synthetic minority over-sampling resampling algorithms (*46*). The random under-sampling approach under-samples the majority classes by randomly picking samples from the majority class (high speed records). The over-sampling approach generates new samples in the classes that are under-presented (low speed records) with replacements. The synthetic minority over-sampling technique (SMOTE) was also used to over-sample the minority classes and under-sample the majority classes. The augmented data sets were used along with the original data set to train the classifiers.

Model Development

Traffic data sets generated from the resampling methods and the original data set were used as an input for classification algorithms. Five-fold cross-validation was used to avoid over-fitting and selection bias. This cross-validation approach generated complementary subsets of data in which a certain portion of data was used to train the algorithms and the remaining subset was used to evaluate the model's generalization. Multiple rounds of cross-validation were performed to assess variability in the model's predictive performance. An average of these results was used to represent the overall performance of the classification models. Hyper-parameter tuning was performed using grid search and randomized search (*47*) methods to further evaluate the models confidence and reliability (*48*). The grid search has been a widely used method for optimizing hyper-parameters while the randomized method is a more recent method that requires less computational resources. The grid search method creates exhaustive combinations of grid parameters to find optimized hyper-parameters. However, the randomized method randomly creates combinations of hyper-parameters based on the distribution of these parameters. After tuning the hyper-parameters, the ANN architecture included three hidden layers with 30, 40, and 50 nodes, respectively. Figure 4 provides a schematic view of the ANN architecture highlighting the input attributes used for the training, hidden layers, and output of the model.

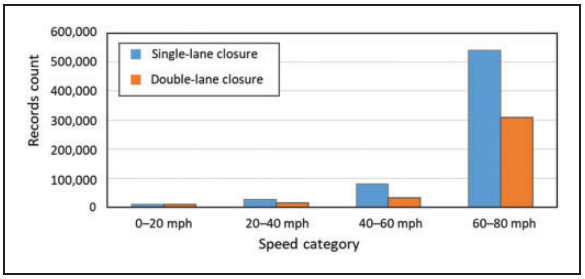


Figure 3. Record count distribution for single and double lane closures.

Model Evaluation

Several measures were used to evaluate the model's performances. Measures used to evaluate prediction for each class were precision, recall, and F1-score. Precision is the number of correctly predicted positive observations divided by the total number of positive observations predicted by the classifier. Recall is the ratio of correctly predicted positive observations to the number of all the positive observations. F1-score is the harmonic mean between precision and recall. In imbalanced data set cases, these metrics can be used to evaluate performance for each class, but they could be misleading as performance measures to evaluate overall performance. For example, when a training set consists of an unbalanced portion for each class label, the predicted result can be biased toward the more frequent class. Consequently, by applying these metrics to test an imbalanced data set, the classifier may be prone to estimate higher accuracy than is realistic.

Therefore, the performance of these models was evaluated using micro and macro average of these metrics (Equations 4–9). Micro and macro averages represent two ways of interpreting prediction performance in multi-class settings. A macro average computes each of these metrics independently for each class and then takes the average, whereas a micro average aggregates the contributions of all classes to compute the average metric (*49*). Overall model evaluation was performed using the macro average of recalls or balanced accuracy (Equation 8) score to address the imbalanced data issue and evaluate the generalization of these models (*50*). Mathematical formulas to calculate each of these measures are provided in the next section. We will use TP_i, FP_i, and FN_i to

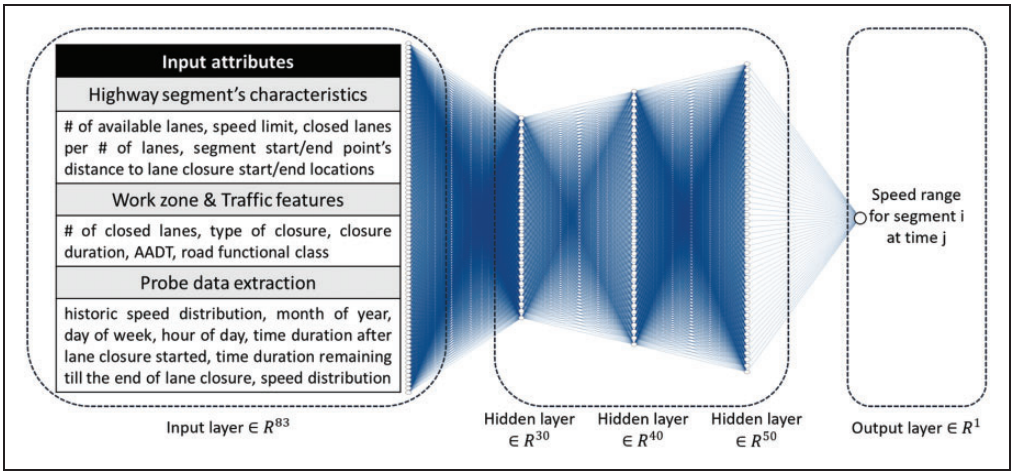


Figure 4. Artificial neural network architecture.

respectively indicate true positives, false positives, and false negatives in the confusion matrix associated with the *i*th class. A true positive is an outcome where the model correctly predicts the target class. A false positive is an outcome where the model incorrectly predicts the target class. Also, a false negative is an outcome where the model incorrectly predicts other than the target class. G represents number of classes. Precision and recall are shown as P and R, respectively.

$$Precision : \frac{TP_i}{TP_i + FP_i} \tag{1}$$

$$Recall : \frac{TP_i}{TP_i + FN_i} \tag{2}$$

$$F1 - Score : \frac{Precision \times Recall}{Precision + Recall} \tag{3}$$

$$MicroPrecision = \frac{\sum_{i=1}^{|G|} TP_i}{\sum_{i=1}^{|G|} TP_i + FP_i} \tag{4}$$

$$MicroRecall = \frac{\sum_{i=1}^{|G|} TP_i}{\sum_{i=1}^{|G|} TP_i + FN_i} \tag{5}$$

$$MicroF1 - Score = 2\frac{P_{micro} \times R_{micro}}{P_{micro} + R_{micro}} \tag{6}$$

$$MacroPrecision = \frac{1}{|G|}\sum_{i=1}^{|G|} \frac{TP_i}{TP_i + FP_i} \tag{7}$$

$$MacroRecall = \frac{1}{|G|}\sum_{i=1}^{|G|} \frac{TP_i}{TP_i + FN_i} \tag{8}$$

$$MacroF1 - Score = 2\frac{P_{macro} \times R_{macro}}{P_{macro} + R_{macro}} \tag{9}$$

Example and Analysis

To illustrate the applicability of this approach, a traffic speed heat map was used to show predicted values compared with actual observations for the same work zone illustrated in the methodology section. Figure 5 provides a visual comparison between the predicted traffic speeds and actual observations. Figure 5*a* shows actual traffic speeds for each segment during the lane closure while Figure 5*b* shows predicted speeds using the ANN model. As shown, the ANN model was able to capture the overall pattern of traffic congestion on this corridor.

Model Performances

Table 1 provides the classification report generated for each of the applied models. Precision, recall, and F1-score metrics are reported based on each class. Micro and macro averages of these metrics are also shown to provide overall performance comparison between these models. As discussed above, balanced accuracy score was chosen to compare these model's performances. Among these models, ANN outperformed the random forest and XGBoost models by reaching up to 85% balanced accuracy. The ANN model used in this study consisted of three hidden layers with 30, 40, and 50 nodes, respectively. Hyperparameters tuned using the grid search method showed better results than the randomized method. The data set

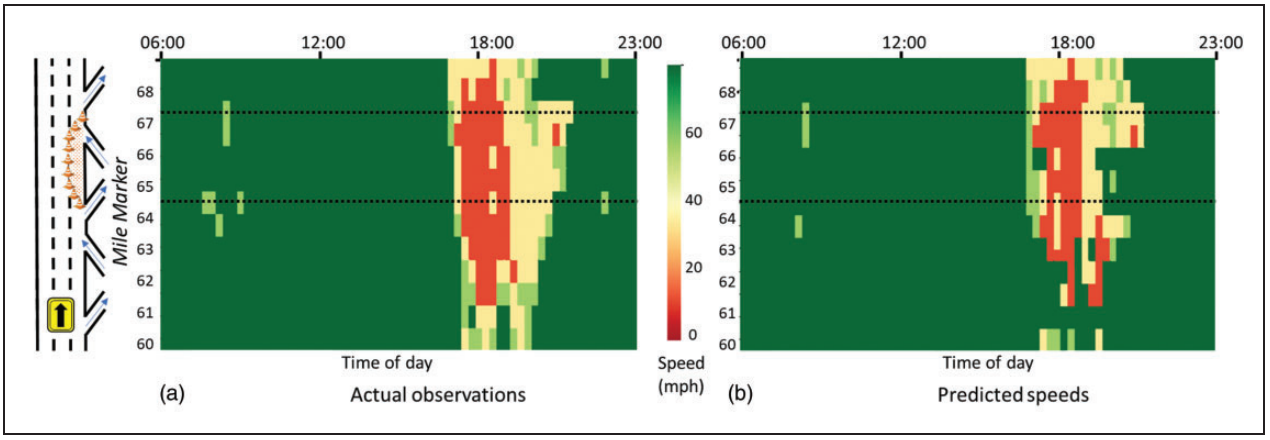


Figure 5. Speed heat maps from (*a*) actual observations and (*b*) predicted speeds.

Table 1. Model Performance Results

		Evaluation metrics			
Model	Speed range (mph)	Precision	Recall	F1-score	Balanced accuracy
Random forest	0–20	0.80	0.76	0.78	0.74
	20–40	0.67	0.67	0.67	na
	40–60	0.66	0.61	0.63	na
	60–80	0.94	0.95	0.94	na
	Macro average	0.76	0.74	0.75	na
	Micro average	0.87	0.87	0.87	na
XGBoost	0–20	0.78	084	0.81	0.79
	20–40	0.69	0.74	0.71	na
	40–60	0.67	0.65	0.66	na
	60–80	0.94	0.94	0.94	na
	Macro average	0.77	0.79	0.78	na
	Micro average	0.88	0.88	0.88	na
Artificial neural network	0–20	0.85	0.88	0.87	0.85
	20–40	0.79	0.80	0.80	na
	40–60	0.77	0.73	0.75	na
	60–80	0.95	0.96	0.96	na
	Macro average	0.85	0.85	0.85	na
	Micro average	0.92	0.92	0.92	na

Note: na = not applicable.

used for training was an augmented data set created by SMOTE resampling technique. The SMOTE resampling technique generated a better synthetic data set compared with both the original and the other augmented data sets created by random under-sampling and over-sampling methods. From each class perspective, the ANN model was able to predict queuing conditions (0–20 mph) with 88% accuracy compared with XGBoost and random forest models with 84% and 76% accuracies, respectively.

A clean and unambiguous way to present the prediction results of a classifier is to use a confusion matrix. Figure 6 shows a confusion matrix for the ANN model. This figure shows that predicted traffic speeds are clustered around the top left to bottom right diagonal, which represents the conditions where predicted records are deviating less from actual observations. Because of recurrent fluctuations in traffic speed, the model was expected not to perform well when traffic speeds were transitioning from high to low speeds; however, congested scenarios (queuing conditions) were identified more consistently.

Discussion and Conclusion

In this study, several machine learning algorithms including XGBoost, random forest, and ANN were

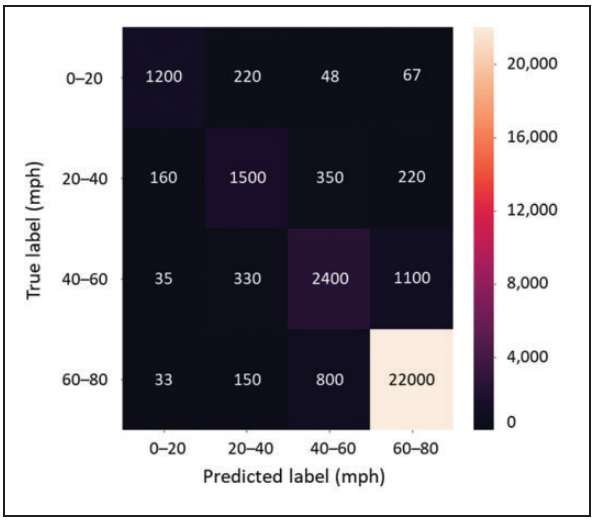


Figure 6. Confusion matrix for artificial neural network (ANN) model.

applied to predict speed range for each highway segment when lane closures were present on interstate highways in the State of Michigan. These models used work zone configuration, roadway geometry, AADT, and historical traffic speeds as inputs. Data from 1,165 historical lane closures from 2014 to 2017 were used to train these classification models. The balanced accuracy score was used for an overall model performance evaluation. The SMOTE resampling approach was used to create the synthetic data set for training. A new mobility baseline was defined using the distribution of historical speeds to represent mobility trends in the absence of hourly traffic volume. The results suggested that ANN outperformed the other models in predicting more accurate speed profiles. This was specially true in cases predicting low speed or queuing conditions. The key advantage of this modeling approach was to use historical traffic speed distributions in the absence of hourly traffic volume to achieve a promising accuracy in predicting the spatio-temporal impact of lane closures on traffic mobility. Some notes from this study are worth mentioning:

- This method is applicable for large-scale implementation to assist planning future lane closure projects.
- As more data become available in future, prediction performance is suspected to improve.
- Models developed in this study are only applicable to predict mobility for Michigan work zones. It is necessary to retrain these models for other locations.
- Including hourly traffic volume and weather information, if available, would help to establish more accurate predictions.
- Agencies need to develop systematic approaches for recording lane closure activities. This will have tangible returns in future modeling practices.
- The choice of speed bins could be different depending on an agency's thresholds and policies. It is recommended for future work to examine unsupervised learning approaches such as clustering algorithms to find the optimal choice of speed bins to maximize prediction accuracy.
- Future research work could also incorporate image processing algorithms to assess heat maps during work zones and predict future scenarios (*51*).
- It is recommended to examine machine learning algorithms to assist in scheduling future work zones. This approach could use these prediction results and recommend best times to close highway lanes minimizing the negative impact on mobility.

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Author Contributions

The authors confirm contribution to the paper as follows: study conception and design: M. Kamyab, S. Remias; data collection: M. Kamyab, S. Remias; analysis and interpretation of results: M. Kamyab, E. Najmi, S. Rabinia; draft manuscript preparation: M. Kamyab, S. Remias, S. Rabinia, J. M. Waddell. All authors reviewed the results and approved the final version of the manuscript.

Declaration of Conflicting Interests

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